

JOSIAH CALVERT

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EDUCATION

University of Nottingham MSci in Physics with Theoretical Physics - First Class Honours <i>Notable grades:</i> Quantum Field Theory 92%, Quantum Dynamics 88%, Classical Fields 82%, Theory Toolbox 76%	Sept 2022 - July 2026
Chellaston Academy A-Level Maths, Further Maths, Physics, Chemistry (A*, A*, A*, A*)	Sept 2015 - July 2022

SKILLS AND INTERESTS

Technical Skills	Python (NumPy, SciPy, Matplotlib), LaTeX, Git
Theoretical Physics	Quantum field theory, general relativity, gauge theory, conformal field theory, gravitational thermodynamics
Mathematical Methods	Differential geometry, tensor calculus, operator algebras, variational and spectral methods, perturbation theory, Fourier analysis, numerical ODE/PDE methods
Research Interests	Quantum gravity, holography and emergent spacetime, AdS/CFT, operator-algebraic approaches to physics, string theory

PROJECTS

Spectral Holography: Emergent Geometry from Entanglement <i>4th Year Project</i>	Sept 2025 - July 2026 <i>Supervisor: Prof Avgoustidis</i>
- Developed and critically evaluated a mathematical framework connecting entanglement and operator-algebraic structures to emergent geometric and gravitational data on finite weighted graphs - Worked with von Neumann algebras, crossed products, modular theory, spectral graph theory and holonomy to construct geometric and gauge-theoretic structures from quantum information - Investigated the relationship between spectral data, entanglement and spacetime geometry, including constructions motivated by holography and gravitational thermodynamics - Applied the framework to model gravitational and gauge-theoretic phenomena, including black-hole entropy and emergent geometric dynamics	
Supersymmetry in Quantum Mechanics <i>3rd Year Project</i>	Jan 2025 - Jun 2025 <i>Supervisor: Prof Saffin</i>
- Produced a comprehensive review paper on supersymmetry, covering supersymmetric quantum mechanics, supersymmetric field theories, Grassmann algebras and superalgebraic structures - Worked through the formalism of supersymmetric quantum mechanics, including factorisation methods, shape invariance, supercharges, partner Hamiltonians and supersymmetric hierarchies - Examined physical applications of supersymmetry, including its role in beyond-Standard-Model phenomenology, collider searches at the LHC, and superstring theory, evaluating the current experimental status and future prospects	
Rotating Wave Approximation in Quantum Coherent Devices <i>Mini-Project</i>	Dec 2025 - Jan 2026
- Numerically investigated the validity of the Rotating Wave Approximation across coupling regimes, benchmarking entanglement entropy predictions against the full quantum Rabi Hamiltonian - Demonstrated RWA breakdown at moderate-to-strong coupling, highlighting practical constraints for quantum sensing protocol design in superconducting qubit platforms	

INDEPENDENT RESEARCH

York-Time Quantum Clocks in Canonical Gravity

Independent Research

2026

Research Manuscript

- Investigated York time as a dynamical variable in canonical general relativity, examining its role in defining physical evolution and addressing the problem of time in quantum gravity
- Developed a quantum-clock framework based on the ADM decomposition and York-time formalism, studying how gravitational constraints shape the emergence of relational quantum dynamics
- Explored connections between canonical quantisation, gravitational observables and emergent notions of time

Hodge-Relative Completions of Characteristic Brane Couplings

Independent Research

2026

Research Manuscript

- Investigated whether restoring the metric to characteristic gauge-theoretic couplings through the Hodge star can produce genuinely new local D-brane interactions
- Derived a representative-dependence criterion showing that Hodge completion is representative-independent precisely on the harmonic locus
- Applied the construction to the D4-brane and proved a leading-order no-go result: the resulting couplings reduce to known DBI, flux-kinetic, or characteristic sectors, identifying the role of extrinsic geometry as the natural next extension

PRESENTATIONS

Effective Field Theories

4th Year Presentation

2026

University of Nottingham

- Presented an introduction to effective field theories, discussing their theoretical foundations, separation of energy scales and applications in modern particle physics

The Higgs Boson

2nd Year Presentation

2024

University of Nottingham

- Presented the theoretical motivation for the Higgs mechanism, the role of spontaneous symmetry breaking and the experimental discovery of the Higgs boson at the LHC

ACADEMIC ENGAGEMENTS

Institute of Physics

- Graduate membership of the Institute of Physics

Royal Society Summer Science Exhibition

- Attended multiple times, participating in seminars and discussions with researchers about current developments in physics

EMPLOYMENT

GP Surgery, Swadlincote

Administrative Role

Sept 2026 - Present

- Digitisation and scanning of confidential patient records, requiring accuracy, organisation and careful attention to detail
- Provided administrative support within a professional healthcare environment, following established procedures and maintaining confidentiality
- Developed skills in data handling, records management, document processing and professional communication

ADDITIONAL INTERESTS

- Six years self-taught guitarist, long-term music collector, amateur astronomer